STATISTICS

CHI-SQUARE TESTS

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Chi-Square Tests

Chi-square tests are applied to test categorical variables in terms of fitting to a distribution, dependency between two groups, or homogeneity between two or more groups. Accordingly, there are three types of chi-square tests:

- Chi-Square Test for Goodness of Fit
- Chi-Square Test for Independence
- Chi-Square Test for Homogeneity

Assumptions of the test:

1. Groups are disjoint.
2. The expected values for each category are at least 5.
3. The observations are always assumed to be independent of each other.

Chi-Square Tests for Goodness of Fit

Consider a frequency distribution containing k classes.

Hypothesis:

H_0: The frequency distribution fits a specified distribution (such as normal distribution, uniform distribution, etc.).

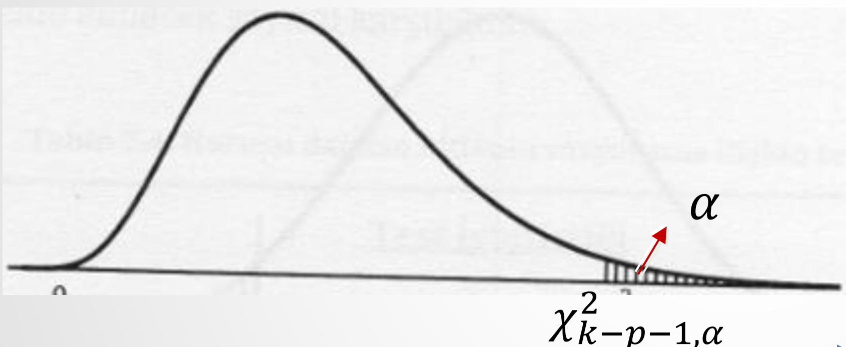
H_a: The frequency distribution does not fit a specified distribution.

Test Statistics:

$$\chi_0{}^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Here, O_i and E_i are the observed and expected frequencies (when H_0 is correct) of the class i, respectively.

Chi-Square Tests for Goodness of Fit



$$\chi_0{}^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

p is the parameter number.

$p = 2$ for normal distribution

$p = 1$ for Poisson distribution

$p = 0$ for uniform distribution

Decision Rule:

If $\chi_0^2 \leq \chi^2_{k-p-1,\alpha}$, fail to reject H_0. Otherwise, reject H_0.

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Example 1: To test a computer program, 1000 random integers have been generated between the interval [0, 9]. The numbers of the generated integers are given in the following table. Accordingly, test whether this computer program generates the random numbers uniformly. ($\alpha = 0.05$).

Integers	*Observed Frequencies*
0	94
1	93
2	112
3	101
4	104
5	95
6	100
7	99
8	108
9	94

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Solution 1:

Integers	*Observed Frequencies*	*Expected Frequencies*
0	94	100
1	93	100
2	112	100
3	101	100
4	104	100
5	95	100
6	100	100
7	99	100
8	108	100
9	94	100

H_0 : The random numbers are generated uniformly.

H_a: The random numbers are not generated uniformly.

α = 0.05

$$\chi_0^2 = \sum_{i=1}^{10} \frac{(O_i - E_i)^2}{E_i}$$

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Solution 1:

$$\chi_0^2 = \sum_{i=1}^{10} \frac{(O_i - E_i)^2}{E_i} = \frac{(94-100)^2}{100} + \frac{(93-100)^2}{100} + \cdots + \frac{(94-100)^2}{100} = 3.72$$

$$\chi^2_{10-1,0.05} = 16.92$$

Since $\chi_0^2 \leq \chi^2_{9,0.05}$, we fail to reject H_0. We have no sufficient proof to reject that the computer program generates the random numbers uniformly with %95 confidence level.

Example 2: In a study on package weights, a sample of size 1000 units was chosen, and its mean and standard deviation were measured as 99 grams and 1.63 grams, respectively. The package weights are classified as given below. Accordingly, test whether the package weights are normally distributed with the given mean and standard deviation at a 99.5% confidence level.

Classes	*Observed Frequencies*
$94 \le X < 96$	100
$96 \le X < 98$	200
$98 \le X < 100$	500
$100 \le X < 102$	150
$102 \le X < 104$	40
$104 \le X < 106$	10

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Solution 2:

Classes	*Observed Frequencies*	*Expected Frequencies*
$94 \le X < 96$	100	31.8
$96 \le X < 98$	200	238
$98 \le X < 100$	500	458
$100 \le X < 102$	150	238
$102 \le X < 104$	40 } 50	31.8 } 34.2
$104 \le X < 106$	10	2.4

H_0: The package weights have a normal distribution with a mean of 99 grams and a standard deviation of 1.63 grams.

H_a: The package weights do not have a normal distribution with a mean of 99 grams and a standard deviation of 1.63 grams.

α = 0.005

$1000\ x\ 0.0318$

$Since\ E_6 < 5$, merge the classes→ O_5 =50, $E_6 = 34.2$

$$P(94<X<96)= P\left(\frac{94-99}{1,63} <z< \frac{96-99}{1,63}\right) = P(-3,067<z<-1,84) = 0,4989-0,4671 = 0,0318$$

$$P(96<X98) = P(-1,84<z<-0,61) = 0,4671-0,2291= 0,238$$

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Solution 2:

$$\chi_0^2 = \sum_{i=1}^{5} \frac{(O_i - E_i)^2}{E_i} = \frac{(100 - 31.8)^2}{31.8} + \cdots + \frac{(50 - 34.2)^2}{34.2} = 195.94$$

$$\chi^2_{5-2-1,0.005} = 10.6$$

$p = 2$: Normal distribution have two parameters, namely μ and σ

Since $\chi_0^2 \geq \chi^2_{2,0.005}$, we reject H_0. The package weights do not have a normal distribution a mean of 99 grams and a standard deviation of 1.63 grams.

Chi-Square Tests for Independence (Single Categorical Variable)

Consider a frequency distribution of a categorical variable that can take k different values.

Hypothesis:

H_0: The categorical variable is independent of the frequencies.

H_a: The categorical variable is dependent on the frequencies.

Test Statistics: $${\chi_0}^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Decision Rule:

If $\chi_0^2 \leq \chi_{k-1,\alpha}^2$, fail to reject H_0. Otherwise, reject H_0.

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Example 3: One-year television sales in a store were observed, and their number by month was found as follows. According to this; test whether the television sales are dependent on the months with a 95% confidence level.

Months	*Observed Freq.*
January	18
February	19
March	25
April	24
May	20
June	18
July	16
August	22
September	30
October	35
November	28
December	21

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Solution 3:

Months	*Observed Freq.*	*Expected Freq.*
January	18	276/12=23
February	19	276/12=23
March	25	276/12=23
April	24	276/12=23
May	20	276/12=23
June	18	276/12=23
July	16	276/12=23
August	22	276/12=23
September	30	276/12=23
October	35	276/12=23
November	28	276/12=23
December	21	276/12=23

Total: 276

H_0: The television sales are independent of months.

H_a: The television sales are dependent on months.

$$\alpha = 0.05$$

$$\chi_0^2 = \sum_{i=1}^{12} \frac{(O_i - E_i)^2}{E_i} = 15.3$$

$$\chi^2_{12-1,0.05} = 19.68$$

Since $\chi_0^2 \le \chi^2_{12,0.05}$, we fail to reject H_0. We have no sufficient proof to reject that the television sales are independent to months.

Chi-Square Tests for Independence (Two Categorical Variables)

Consider a $k \times c$ contingency table (frequency distribution of two categorical variables that have k rows and c columns).

Hypothesis:

H_0: The categorical variables are independent.

H_a: The categorical variables are dependent.

Test Statistics:

$$\chi_0^2 = \sum_{i=1}^{k} \sum_{j=1}^{c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}}$$

Decision Rule:

If $\chi_0^2 \leq \chi^2_{(k-1)\times(c-1),\alpha}$, fail to reject H_0. Otherwise, reject H_0.

Example 4: There are three different retirement plans in a company. Test whether there is an association between these retirement plan preferences and people's hourly and salaried work, i.e., whether the two criteria are independent at the significance level $\alpha = 0.05$.

	Retirement Plan 1	Retirement Plan 2	Retirement Plan 3	Total
Salaried Work	160	140	40	340
Hourly Work	40	60	60	160
Total	200	200	100	500

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Solution 4:

Observed	**Retirement Plan 1**	**Retirement Plan 2**	**Retirement Plan 3**	**Total**
Salaried Work	160	140	40	340
Hourly Work	40	60	60	160
Total	200	200	100	500

Expected	**Retirement Plan 1**	**Retirement Plan 2**	**Retirement Plan 3**	**Total**
Salaried Work	$200\frac{340}{500}=136$	$200\frac{340}{500}=136$	$100\frac{340}{500}=68$	340
Hourly Work	$200\frac{160}{500}=64$	$200\frac{160}{500}=64$	$100\frac{160}{500}=32$	160
Total	200	200	100	500

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Solution 4:

H_0: The retirement plans and working types are independent.

H_a: The retirement plans and working types are dependent.

α = 0.05

$$\chi_0^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} = \frac{(160-136)^2}{136}+\frac{(140-136)^2}{136}+\cdots+\frac{(60-32)^2}{32} = 49.63$$

$$\chi^2_{(2-1)\times(3-1),0.05} = 5.99$$

Since $\chi_0^2 \geq \chi^2_{2,0.05}$, we reject H_0. The retirement plans and working type are dependent.

Chi-Square Tests for Homogeneity

Consider a $k \times c$ contingency table (frequency distribution of a categorical variable with k values given for c different populations).

Hypothesis:

H_0: The distribution of the categorical variable is homogeneous for all populations.

H_a: The distribution of the categorical variable is not homogeneous for all populations.

Test Statistics:

$$\chi_0^2 = \sum_{i=1}^{k} \sum_{j=1}^{c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}}$$

Decision Rule:

If $\chi_0^2 \leq \chi^2_{(k-1)\times(c-1),\alpha}$, fail to reject H_0. Otherwise, reject H_0.

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Example 5: To examine whether the frequency of going to the theater is the same for students studying in different faculties; 40 students from the Science Faculty, 42 students from the Faculty of Economics, and 38 students from the Communication Faculty were asked about the frequency of going to the theater. Analyze whether the frequency of going to the theater is homogeneously distributed according to the data in the table. ($\alpha = 0.05$)

	Science	Economics	Communication	Total
Frequently	6	14	16	36
Rarely	25	22	17	64
Never	9	6	5	20
Total	**40**	**42**	**38**	**120**

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Solution 5

Observed	Science	Economics	Communication	Total
Frequently	6	14	16	36
Rarely	25	22	17	64
Never	9	6	5	20
Total	**40**	**42**	**38**	**120**

Expected	Science	Economics	Communication	Total
Frequently	$40\frac{36}{120} = 12$	$42\frac{36}{120} = 12.6$	$38\frac{36}{120} = 11.4$	36
Rarely	$40\frac{64}{120} = 21.3$	$42\frac{64}{120} = 22.4$	$38\frac{64}{120} = 20.3$	64
Never	$40\frac{20}{120} = 6.7$	$42\frac{20}{120} = 7$	$38\frac{20}{120} = 6.3$	20
Total	**40**	**42**	**38**	**120**

Solution 5:

H_0: The frequencies of going to theater are distributed homogeneously for given faculties.

H_a: The frequencies of going to theater are distributed inhomogeneously for given faculties.

α = 0.05

$$\chi_0^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij}-E_{ij}\right)^2}{E_{ij}} = \frac{(6-12)^2}{12}+\frac{(14-12.6)^2}{12.6}+\cdots+\frac{(5-6.3)^2}{6.3} = 7.4$$

$$\chi^2_{(3-1)\times(3-1),0.05} = 9.49$$

Since $\chi_0^2 \leq \chi^2_{4,0.05}$, we fail to reject H_0. There is no sufficient proof to reject that the frequencies of going to theater are same for given faculties.